Sustainable Travel Planner

Project Documentation

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Course: CMSC 207 Web Programming & Development

1. Overview

The **Sustainable Travel Planner** is a PHP-based web application that helps users plan environmentally friendly trips and commute segments. It allows users to set up **"Destination Goals"** and map out transportation modes to reach them, while computing an **Eco-Score** and an overall **Eco-Rating** based on the carbon emissions of selected transport modes.

2. Core Features

User Authentication

- Secure registration (`register.php`) and login (`login.php`) systems
- Session-based user tracking

Destination Goals

- Create and manage travel destinations (`destination_form.php`, `destination_detail.php`)
- Destinations are displayed on the dashboard (`index.php`) with optional cover photos

Trip Segments & Planning

- Map out individual travel legs in `planner.php`
- Add transport modes (e.g., Walking, Bike, Train, Bus, Flight)
- Define whether the trip is going *to* the destination or within the destination

Eco-Rating System

- Transport modes have emission factors
    - Example: Walking and Bike = **0.00 (eco-friendly)**
- `travel_calculate_eco_rating` evaluates sustainability (scale of **1.0 to 5.0**)
- Based on:
    - Average carbon emissions
    - Ratio of eco-friendly transport usage

Photo Management

- Upload and manage destination images

3. Tech Stack

- **Frontend:** HTML5, CSS3 (`style.css`, responsive grid layout)
- **Backend:** PHP 8+
- **Database:** MySQL (`eco_travel`) via `db.php`
- **Host Environment:** XAMPP (Apache + MariaDB)

4. File Structure

- `index.php` – User dashboard displaying destination goals
- `login.php`, `register.php` – Authentication modules
- `logout.php` – Ends user sessions securely
- `planner.php`, `trip_edit.php` – Travel planning forms
- `destination_*.php` – Destination management modules
- `trip_segment_helper.php`, `trip_handler.php` – Business logic and eco-calculations
- `db.php` – Database connection handler
- `eco_travel.sql` – Database schema and seed file

5. Usage / Setup Guidelines

1. Copy the project folder **`sustainable_travel`** into:
   `C:\xampp\htdocs`
2. Open the XAMPP Control Panel and start:
   - Apache
   - MySQL
3. Go to:
   `http://localhost/phpmyadmin`
   - Create a database named **eco_travel**
   - Import the file **eco_travel.sql**
4. Open your browser and visit:
   `http://localhost/sustainable_travel`
5. Register a new user account and start planning sustainable trips